A Comparative Study on the Detection of Image Forgery of Tampered Background or Foreground

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***Abstract*— Recently tampering and manipulating of images has become easier with advanced image editing applications. The process of making a composite image by combining image fragments from different images is called image splicing. One of the common types of image splicing forgery is to change the whole background of the images for different purposes. However, very few studies in the literature directly address the tampered background or foreground detection. We present a comparative study and analysis of image splicing detection methods on the detection of image forgery of tampered background or foreground. First, we create a novel dataset contains both tampered and non-tampered images. Using the dataset, we conduct a comparison of the state-of-the-art image splicing detection methods. The experimental results indicate that existing image splicing detection methods are not sufficient for detecting background or foreground forgeries, and new methods should be developed.**

Keywords—background forgery, image forgery, image splicing detection.

I. Introduction

Image manipulation has become ubiquitous in today's world through the widespread use of advanced image editing applications. Image manipulation techniques can be used for image enhancement as well as for image forgery. The most widely used image forgery techniques are copy-move and image splicing techniques [1]. In Copy-move forgery, image regions corresponding to individual surfaces, objects, or parts of objects are copied and pasted onto the same image. On the other hand, image splicing methods combine image regions taken from multiple images to create a forged image [2].

One of the most common digital image forgery techniques is to change the background of the images. The background of the images can be changed for different purposes. Such as, entertainment purposes to produce a photo of a person that looks like the photo were taken in a place where he/she has never been. It is also used to exhibit and highlight subjects for advertisements. In video conference calls, people have started to use virtual backgrounds to prevent information about their location from being visible to other peoples in the call. Increasing the requirement for background changing processes for different purposes has made the use of applications specialized for this process widespread.

Digital image forgery techniques are classified as active and passive methods. Active methods require an additional data such as digital watermarking or digital signature for image verification [3,4]. However, in passive methods, there is no requirement on the additional data. Passive image forgery detection methods first extract features from images or image data (e.g., noise, illuminance, EXIF data) and then classify these features using particular classifiers [5-7]. The main idea behind these techniques is that although digital forgeries may leave no visual clues on images, they would unavoidably alter the underlying statistical data of the image. In this context, He et al. [8] proposed an image splicing detection method using Markov features in discrete cosine transform (DCT) and Discrete Wavelet Transform (DWT) domain. Rao et al. [9] used the Support Vector Machine (SVM) to classify CNN features to detect image splicing. Wu et al. [10] used the illuminant color inconsistency to detect image splicing.

Recently, deep learning-based methods have become popular in image forgery detection due to their state-of-the-art performance. Zhou et al. developed a method that uses a two-channel Faster R-CNN network in which one channel uses RGB image, and the other uses the Steganalysis Rich Model (SRM) noise stream to detect different types of image forgeries [11]. Wang et al. proposed a weight combination strategy for image splicing detection. This method depends on three different types of features on reasonable ratio using combination strategy [12]. Pomari et al. proposed a method that first extracts features from RGB images and illumination maps and then transfers them to the classifier to detect image splicing [13]. Cozzolino et al. proposed a deep learning model that extracts noise from images and uses it for image forgery detection [14]. Huh et al. developed a deep learning-based method that uses image metadata for image splicing detection. In this study, they proposed a Siamese network which trained with patches from different images to determine the same or different EXIF data to detect image spliced area in an image [15]. Wu et al. developed a deep learning-based image forgery detection and localization solution called ManTraNet, which consists of image manipulation trace feature extractor and local anomaly detection networks [16].

Background or foreground manipulation is one of the ways of image splicing. Although many studies have been carried out on image splicing detection, studies on background or

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foreground forgery are very limited. As far as we know, the only study that directly targets this issue is an active method proposed by Yang et al. [17]. This method first uses spectral matting to obtain foreground and background regions of the image, then embeds two different watermarks into foreground and background, respectively, using a watermarking technique that employs DWT, DCT, and singular value decomposition (SVD). Lastly the watermarks are removed from the background and foreground. Using these watermarks background or foreground forger is detected.

In this work, we conduct a comparative study of image forgery detection methods to evaluate the performance of these methods in images with completely changed foreground or background. First, we construct a novel dataset contains non-tampered images and tampered images with changed backgrounds. Then, we made a comparison of six state-of-the-art image splicing detection methods on this dataset. We conduct both visual and numerical evaluations to analyze the performance of these methods on background or foreground forgery.

The contribution of this paper is twofold:

- We construct a dataset for evaluating image forgery detection algorithms. Our dataset contains both tampered and non-tampered images. To the best of our knowledge, this is the first dataset for foreground or background forgery.
- We evaluate the three state-of-the-art deep learning-based and three feature-based image forgery detection algorithms on the dataset.

The rest of the paper is organized as follows: In section 2, a novel image dataset and compared image forgery detection methods are introduced. In section 3, the performances of the six image forgery detection methods are compared on the proposed dataset. Finally, section 4 concludes the study.

II. Experimental Settings

A. Image Dataset

This study aims to evaluate the performance of state-of-the-art image splicing detection methods on background forgery detection tasks. To this end, we have first collected one thousand images from Flickr that contain at least one person in the foreground. We have initially chosen five hundred images among them. We then segmented these images into human and background classes and changed the background. In order to ensure the integrity of the human class in the images, the clothes of the human or the objects in their hand were also included in the human class. We have used different backgrounds to prevent the models that use the dataset for training from learning fake backgrounds. We have labeled images with a changed background as tampered while unaltered images as non-tampered. Fig. 1 shows an example image from the dataset. The original image and background image are illustrated in Fig. 1(a) and (b), respectively. Fig. 1(c) demonstrates the reference mask representing the person class for Fig. 1(a). Fig. 1(d) shows the tampered image using the background given in Fig. 1(c).

B. Evaluated Algorithms

Image splicing detection methods can consider various visual and statistical features of the images to detect image forgeries. In the literature, both feature-based and deep learning-based image splicing methods are proposed. We selected three deep learning-based and three feature-based state-of-the-art image splicing methods to evaluate their performance on background forgery detection using the created dataset [14-20]. This section briefly introduces the compared methods.

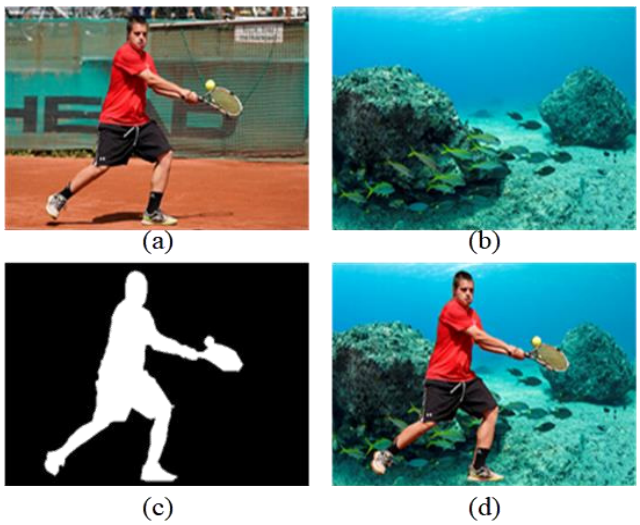

Fig. 1. An example of background tampered image: (a) original image; (b) background image; (c) reference mask; (d) tampered image.

1) Noiseprint Model [14]: In the Noiseprint algorithm noise data which called Noiseprint are used to perform to detect and localize image manipulations. Firstly, the noise of the input image is obtained by performing the CNN based residual noise extraction process. The CNN based denoiser training with patches of image and their noise extracted from AWGN pattern. For training phase, the weights are optimized using distance between model predictions and the patches of noise. To localize the tampered image regions, each region part is associated with its noise feature vector. Then, the noise vectors are used to recognize image manipulation by giving two different noise parts belonging to same image to a Siamese network at the same time.

2) **Learned Self-Consistency-Based Method (SC) [15]:** This model used EXIF metadata information to detect whether the image is authentic or not. In this method a Siamese network is trained to determine whether two image patches from different images have similar EXIF metadata or not. They used a Siamese network to predict consistency of image patches can be used on any image without further information. It works by predicting a pair of image patches to determine whether they are consistent or not with each other. The Siamese network is used to predict probability that a pair of 128×128 image patches. The network uses Resnet50 as backbone

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network which produces 4096 features. The features are concatenated and passed to a MLP (Multi-Layer Perceptron Network). Then the feature is classified by simple classifier at the end of the model. The model produces a heatmap highlights forged region for input image.

3) ManTraNet [16]: It aims to detect 385 different type image manipulations by dividing the image into blocks of random sizes. This model consists of feature extractor and local anomaly detection network (LADN) respectively. They were determined by carrying out performance experiments. In feature extractor, features are extracted from image and passed through LADN. LADN firstly determines dominant feature, then the process of calculating how different the feature of the region under the dominant feature is calculated. Then LADN the localize and predict forgery type of forged parts of image.

4) Splicebuster [18]: Splicebuster methods explore and indicate the performance of SRM features to determine whether tampered or pristine regions in image. Three main steps for this study: Feature extraction for capture the traces from locally in-camera processing. Computing mean vector and covariance matrix for test images. Using the calculated statistical data, finding local differences from the feature indicates possible image manipulation.

5) First Digit Features Based Method (FDF) [19]: A method proposed for JPEG double compression analysis. The method uses first digit features which extracted from the DCT coefficients. First digit permits to estimate number of compressions an image has suffered using DCT divergence. When an image given the method firstly, DCT coefficient are extracted for each 8×8 block, and first 9 spatial frequencies is selected without double compression (DC). Secondly a histogram is created with every spatial frequency using only 2, 5, 7 from first digits. Thus, size of feature vector is decreases without any crucial loss. Then a SVM is classified by the feature vector. The tampered regions are detected by classifying image blocks as single and double JPEG compressed blocks.

6) Color Filter Array-Based Method (CFA) [20]: Dirik et al. proposed a local and global image tamper detection method consist of computing single feature and a threshold-based classifier. The method is applicable to a variety of tampered operations such as splicing, retouching etc. The method exploited Color Filter Array (CFA) Interpolation for image authentication. In this method two type feature calculated. First one is CFA pattern number estimation. The CFA pattern was calculated like that, to identify the CFA pattern of an input image, the input was re-interpolated with a number of candidate CFA patterns. Second feature is CFA based noise analysis. CFA demosaicing artifacts could be measured to look at sensor noise power changes all the image. CFA demosaicing artifacts can be measured using the ratio of noise variances of interpolated and un-interpolated pixels. If this ratio is near to 1, given image is thought to be tampered. Table I lists the details of the evaluated methods.

TABLE I. EVALUATED IMAGE SPLICING DETECTION ALGORITHMS

Method	Details of Compared Methods		
	Technique	***Additional Information***	***Reference***
Noiseprint	Siamese Network	Noise Data	[14]
SC	Siamese Network	EXIF Data	[15]
ManTraNet	Fully Convolutional Neural Network	-	[16]
Splicebuster	Feature Based Noise Component	-	[18]
FDF	DCT and SVM	-	[19]
CFA	Feature Based CFA Component	CFA	[20]

III. EXPERIMENTAL RESULTS

A. *Quality Metrics*

We utilize pixel accuracy, F1 score, precision, recall, and Intersection over Union (IoU) quality metrics, which are commonly used in the image segmentation evaluation to compare the image splicing detection methods. This section introduces the quality metrics that are employed in experiments [21].

1) Pixel Accuracy

Pixel accuracy is a widely used evaluation metric for semantic segmentation. Pixel accuracy is calculated by taking the ratio of correctly predicted pixels to all pixels. The following equation is used to calculate pixel accuracy:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{1}$$

where, true positive (TP) is the correctly predicted number of pixels that belong to the human class in this task. Similarly, true negative (TN) is the correctly predicted number of pixels that belong to the background class. False positive (FP) is the outcome where the method incorrectly predicts the human class. False negative (FN) is the outcome where the method incorrectly predicts the background class.

2) Precision

The precision metric calculates how many of the predicted pixels belonging to the person class were correctly predicted. Eq. (2) is used to calculate the precision metric:

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

3) Recall

Recall metric calculates what proportion of actual human class pixels is correctly predicted. Eq. (3) is used to calculate the recall metric:

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

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4) Jaccard Index (IoU)

The Jaccard index, also known as Intersection over Union (IoU), is an evaluation metric for segmentation tasks that quantify the proportion of overlap between prediction and label. IoU metric is calculated using Eq. (4):

$$IoU = \frac{label \cap prediction}{label \cup prediction} = \frac{TP}{TP + FP + FN} \tag{4}$$

5) F1 Score (Dice)

The F1 score, also called the Dice metric, is a segmentation evaluation metric related to the IoU. F1 score is the harmonic mean of precision and recall. F1 score is calculated using Eq. (5):

$$F1 = \frac{2 \times precision \times recall}{precision + recall} \tag{5}$$

B) Evaluation and Analysis

In this section, six state-of-the-art image splicing detection methods were evaluated both quantitatively and visually [22]. Among them Noiseprint and Splicebuster methods produce a heatmap that demonstrates forged regions on the input image as a result. On the other hand, SC, ManTraNet, FDF and CFA methods give a gray-scale map that shows the forged regions on the input image as a result. Firstly, the heatmap files are converted to gray level images. Outputs of the FDF and CFA methods scaled up to reference image sizes using the nearest neighbor interpolation algorithm. To make a fair comparison we converted each gray-scale image output to binary mask map using Otsu method. Table II gives the quantitative comparison of the image splicing detection methods in terms of different objective metrics.

TABLE II. QUALITY METRICS RESULTS OF THE COMPARED METHODS ON THE PROPOSED DATASET

Method	**Evaluation Metrics**				
	Accuracy	***Precision***	***Recall***	***IoU***	***F1 Score***
Noiseprint	0.392	0.143	0.234	0.111	0.162
SC	0.655	0.180	0.211	0.126	0.174
ManTraNet	**0.830**	**0.348**	0.121	0.108	0.164
Splicebuster	0.193	0.163	**0.478**	**0.160**	**0.226**
FDF	0.279	0.170	0.428	0.158	0.223
CFA	0.385	0.149	0.288	0.117	0.176

As seen in Table II, the deep learning-based methods (Noiseprint, Self-Consistency, ManTraNet) have been more successful than the feature-based algorithms in terms of Accuracy and Precision metrics. However, the feature-based algorithms have produced better results than the deep learning-based algorithms for the Recall, IoU and F1 Score metrics.

In order to make the comparison more accurate, the visual results were also examined. Fig. 2 shows the visual comparison of the evaluated methods on a tampered image selected from the proposed dataset. The tampered image and the ground truth binary mask are given in Fig. 2(a-b), respectively. Fig. 2(c-h) shows the predicted masks produced by the compared methods. When Fig. 2 is examined, it can be easily seen that deep learning-based methods (see fig. 2(c-e)) produces more meaningful results than the feature-based methods (see Fig. 2(f-h)). Among the deep learning-based algorithms, ManTraNet produces the best result. Since ManTraNet has more complex structure than the others, it can be said that deep learning-based methods with a more complex architecture are required for image forgery detection.

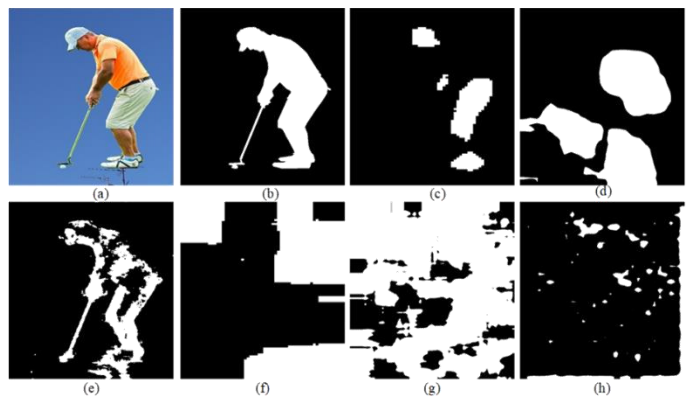

Fig. 2. Visual comparison of the compared methods: (a) original image; (b) reference mask; (c) Noiseprint; (d) SC; (e) ManTraNet; (f) Splicebuster; (g) FDF; (h) CFA.

In this section, a comparison is also made on images whose background is not tampered namely authentic images. It was observed that the compared methods produced results as if there was a forgery in the authentic images. Fig. 3 shows a visual comparison of the evaluated methods on an authentic image. Fig. 3(a-b) show an authentic image and ground truth binary mask. Fig. 3(c-h) demonstrate the forgery detection results of the compared methods on the input image. As can be seen that in Fig. 3(c-e), the deep learning-based methods incorrectly identified the person in the foreground as tampered. It can be concluded that the deep learning-based methods do not perform well in authentic images as opposed to the tampered images.

On the other hand, Noiseprint and Splicebuster methods, which used noise data produced better results than the other methods (see Fig 3(c, f)). So, it can be inferred that; the noise information has a crucial contribution to detect background or foreground tampering.

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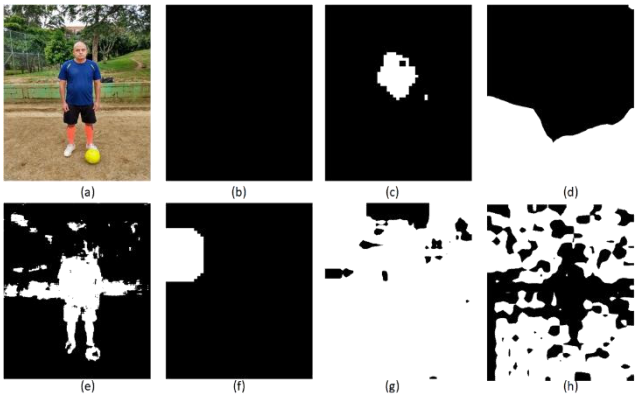

Fig. 3. An example of compared methods results on authentic image: (a) original image; (b) reference mask; (c) Noiseprint; (d) SC; (e) ManTraNet; (f) Splicebuster; (g) FDF; (h) CFA.

IV. Conclusion

In this paper, we conducted a comparative study of image forgery detection methods for background or foreground forgery detection. We first created a novel dataset that contains 1000 images. We compared six state-of-the-art image forgery detection methods, on our dataset. The experimental results showed that using additional information to detect image forgery, has a positive effect on success. We attribute this to even if the forgery does not leave a visual clue in the image, it affects the noise of the image. And last but not least, we observed that the compared methods was not adequate to detect both authentic and tampered image and new methods should be developed in this issue.

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